P-S-576E
August 1, 1978
SUPERSEDING
Fed. Spec. P-S-576D
March 16, 1966

FEDERAL SPECIFICATION

SOAP, GRIT (CAKE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a pumice soap in cake form suitable for removing stubborn dirt from the hands.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-B-636 - Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8520

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Laws and Regulations:

21 CFR 1 - Regulations for the Enforcement of the Food, Drug and Cosmetic Act and the Fair Packaging and Labeling Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Finished product. The finished product shall be a mixture of soap and finely ground pumice in compressed cake form. The grit soap cakes shall be uniform in size, homogeneous in appearance and texture, firm and not crumbly, and free from any gouges, holes, breaks, cracks or chipped surfaces. The grit soap shall be odorless or mildly perfumed. Any dye used shall be approved for cosmetic use by the Food and Drug Administration.

3.2 Prohibited materials. The finished product shall not contain sugar, when tested as specified in 4.3.5. The finished product shall not contain organic matter such as starch, corn meal, flour or sawdust, when tested as specified in 4.3.1.

3.3 Chemical and physical requirements. The finished product shall conform to the chemical and physical requirements of table I when tested as specified in 4.3.

TABLE I. Chemical and physical requirements

Requirement	Minimum	Maximum
Moisture and volatile matter at 105°C (percent)	-	25
Alkaline salts (as Na_2CO_3, percent)	-	1.0
Free alkali (as NaOH, percent)	-	0.1
Free acid (as oleic acid, percent)	-	3.0
Matter insoluble in water (percent)	22	35
Anhydrous soda soap (percent)	35	-
Acid number of mixed fatty acids in soap (mg KOH/g fatty acid)	212	-
Rosin (percent)	-	3.0
Unsaponified matter (percent)	-	0.3
Matter insoluble in water retained on		
No. 100 US Standard Sieve (percent)	-	10.0
No. 200 US Standard Sieve (percent)	10.0	-

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3.4 Cake weight. The cake weight shall be not less than 4.5 ounces nor more then 5.5 ounces. The cake weight shall be not less than that specified in the contract (see 4.2.2 and 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

4.2 Quality conformance inspection.

4.2.1 Examination of preparation for delivery. Packaging, packing and marking shall be examined to determine compliance with the requirements of section 5. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, Acceptable Quality Level (AQL) 4.0 expressed in terms of percent defective. The sample unit shall be one shipping container fully prepared for delivery.

4.2.2 Examination of the grit soap. The grit soap shall be examined for compliance with the requirements of 3.1 and 3.4. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level I, AQL of 2.5 percent defective. The sample unit shall be one cake of grit soap.

4.2.3 Testing of the grit soap. The grit soap shall be tested as specified in 4.3 for conformance with the requirements of section 3. Seal the sample in an airtight container. Exercise care to prevent loss or absorption of moisture during sampling and storage of the sample. The lot shall be rejected if the sample fails to pass any of the tests. In case of failure, report description of failure or numerical point of failure, as applicable.

4.3 Test methods.

4.3.1 Presence of organic matter. Place one gram of the matter insoluble in water obtained in 4.3.3.3 into a porcelain crucible. Heat the crucible and contents strongly with a gas flame. Any charring or burning indicates the presence of organic matter.

4.3.2 Sieve tests. Dry the sieves in an oven at 105°C for 1 hour, cool to room temperature and weigh to the nearest mg. Weigh a 100 g sample of the grit soap to the nearest mg, transfer to a beaker, add 200 ml water and heat on a steam bath for 1 hour to dissolve the soluble matter. Nest the sieves in order from the coarsest on the top to the finest on the bottom. Pour the material from the beaker through the sieves. Rinse the beaker with hot water until all material has been transferred from the beaker to the sieves. Wash the material on the sieves with hot water. If the material forms lumps or aggregates on washing, a camel's-hair brush may be used to break up the lumps. Continue washing until no more material passes through the top sieve. Check by removing the top sieve and washing with 200 ml hot water, collecting the water in a 400 ml beaker. Stir the water in the beaker vigorously and place the beaker on a black surface. If particles are seen collected about the middle of the bottom of the beaker, wash all material from the beaker into the remaining sieves and wash the original sieve with another 200 ml of hot water, again catching the water in the 400 ml beaker. Continue as above until no more particles are seen collected about the middle of the bottom of the beaker. Dry the sieve and residue to constant weight in an oven at 105°C and weigh to the nearest mg. Subtract the tare weight of the sieve to determine the weight of material retained on the sieve. Repeat the above procedure until the material that is retained on each sieve has been determined. Using the result of 4.3.3.3, calculate how much matter insoluble in water was contained in the 100g sample of grit soap. Calculate the percent of that matter insoluble in water which was retained on the sieves.

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4.3.3 Analysis. Analyze the grit soap for the following properties by the referenced sections of ASTM D 460.

4.3.3.1 Matter volatile at 105°C - section 11.

4.3.3.2 Free alkali or free acid - section 18.

4.3.3.3 Matter insoluble in water - section 19.

4.3.3.4 Alkaline salts - section 20.

4.3.3.5 Anhydrous soap - sections 21-22.

4.3.3.6 Unsaponified matter - sections 30-36.

4.3.3.7 Acid number of fatty acids - sections 45-46.

4.3.4 Rosin. If rosin is indicated by the qualitative test in 4.3.3.1, determine quantitatively as specified in 4.3.3.2.

4.3.4.1 Qualitative test for rosin (Liebermann-Storch). Transfer 1 to 2 milliliters (ml) of the fatty acids prepared in section 42 of ASTM D 460 to a test tube, treat with 5-10 ml of reagent grade acetic anhydride, and warm on a steam bath until fatty acids are dissolved. Cool to room temperature. After cooling, pour 1-2 ml into a white porcelain dish and allow a drop or two of sulfuric acid (sp. gr. 1.53) to run down the side of the dish. (Sulfuric acid of sp. gr. 1.53 may be prepared by adding 34.7 ml of concentrated sulfuric acid (sp. gr. 1.84) to 35.7 ml of distilled water.) If rosin is present, a fugitive violet coloration changing to a brownish tinge is immediately produced at the margin of contact of the reagents. This test can be made on a smaller sample on a spot plate. The tests should be checked with a sample of pure stearic acid to which 3 percent of rosin has been added. If rosin is present, proceed with 4.3.4.2.

4.3.4.2 Quantitative test for rosin. Test as specified in ASTM D 460, sections 37-40 - Rosin (McNicoll Method).

4.3.5 Sugar.

4.3.5.1 Reagents.

4.3.5.1.1 Fehling solution. Mix equal volumes of (a) and (b) solutions immediately before use.

(a) Cupric sulfate solution. Dissolve 35 grams of cupric sulfate ($CuSO_4.5H_2O$) in water and dilute to 500 milliliters.

(b) Alkaline tartrate solution. Dissolve 173 grams of Rochelle salt ($NaKC_4H_4O_6.4H_2O$) and 50 grams of sodium hydroxide in water and dilute to 500 milliliters.

4.3.5.2 Procedure. Make 10 ml of a 20 percent soap solution neutral to litmus paper with 1N HCl. After neutralization, add 3 more drops of 1N HCl. Heat on a steam bath for 15 minutes, cool, filter from grit and fatty acids and make neutral to litmus paper with 1N NaOH. Add 5 drops of the filtrate to boiling Fehling's solution and continue boiling for 2 minutes. The formation of red cuprous oxide indicates the presence of sugar.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. One hundred cakes of grit soap shall be packed in a box conforming to PPP-B-636, grade V3c, or V3s. The box shall be provided with top and bottom pads cut 1/4 inch less than the length and width dimensions and made of the same material as the container. The box shall be provided with a case liner fabricated of polyethylene film not less than 0.002 inches thick. The liner shall be sealed to produce moisture barrier seams. Excess air shall be expelled prior to heat sealing or folding of the closure. The box shall be closed, waterproofed and banded in accordance with the appendix of PPP-B-636.

5.1.2 Commercial. The grit soap cakes shall be packed in accordance with normal commercial practice in fiberboard boxes to insure delivery at destination, to provide for redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification and Uniform Freight Classification.

5.2 Marking. Marking shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The grit soap covered by this specification is intended for use in the removal of stubborn dirt from hands.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procuremtn documents:

(a) Title, number, and date of this specification.
(b) Packing required (see 5.1).
(c) Marking required (see 5.2).
(d) Quantity required.

6.3 Bid evaluation. Bids should be evaluated on the basis of price per ounce of soap provided. Bidders shall state the weight of the cake they will provide and the net weight of the soap in the shipping container. The cake weight offered becomes part of any resulting contract. Bid price per ounce of soap is determined by dividing the bid price per shipping container by the net weight of soap in the shipping container.

6.4 The Department of Defense has waived coordination of this specification until further notice.

MILITARY INTEREST

User Activity:

Army - GL

PREPARING ACTIVITY:

GSA - FSS

CIVIL AGENCY COORDINATING ACTIVITIES

GPO
GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 60 cents each.

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